Tut2

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Part I
UFD and prime factorization.

1 Definitions

- **Definition:** A ring R is a PID if

1. R has no zero divisors (integral domain)
2. Every ideal I is of the form (r) for some $r \in R$.

- We call (1) the trivial ideal.

1.0.1 Proposition

- $\mathbb{Z}$, $k[X]$ are PID's.

1.0.2 Proof

- Given $I \subset \mathbb{Z}$, consider $i \in I$ with $|i|$ minimal. We claim $I = (i)$. Indeed, any
$$n = qi + r$$
with $|r| < i$ (Draw picture). Thus if $n \in I$, then $n - qi \in I$, so r has to be zero.

- Given $I \subset k[X]$, consider $i \in I$ with $deg(i)$ minimal....

1.0.3 Example

- Given $i, j \in \mathbb{Z}$, then (i, j) is principal, and the proof gives an algorithm to find $d = gcd(i, j)$, that's the Euclidean algorithm.

1.0.4 Lemma

- If R is a PID and p irreducible, then (p) is a prime ideal- i.e., if $p|ab$, then it divides either a or b.
- **Proof:** Suppose that $p = ab$ and $p \nmid a$, we want to express that p, a are coprime. Then $(g) = (p, a) \subsetneq (p)$, so $g \mid p$, but p is irreducible, and $p \nmid a$, so g must be a unit. That's coprimeness! Thus $\exists \alpha, \beta$ with

  $$\alpha a + \beta p = 1$$

  multiplying by b we obtain

  $$\alpha ba + \beta bp = \alpha p + \beta bp = b$$

  and so $b \in (p)$.
- **Philosophy of the proof:** You see that the correct definition of coprime is a **constructive one.** Good definitions are not negations "p doesn't divide a", but affirmations of data ($\alpha a + \beta p = 1$) that may prohibit other things.

2 Theory- PID is UFD

2.0.1 Combinatorial lemma

- If T is a tree of finite degree at each vertex that doesn't contain any infinite path, then T is a finite tree.
- **Proof sketch:** Suppose by contradiction T has infinitely many vertices. Then by induction, we can find a compatible family of paths P_k of length k which lead to a tree with infinitely many vertices. Then $(P_1, P_2, ...)$ is an infinite path.

2.0.2 Proposition

Any ring R that is a PID is UFD

Proof:

- First, we show any $a \in R$ has a factorization to irreducibles. If a is not irreducible, then $a = bc$. Note $(b), (c) \supset (a)$. If $(b) = (a)$, then

  $$a = ab'c \to 1 = b'c \to c$$

  is a unit. Thus we may assume $(b) \supsetneq (a)$. We can draw a tree corresponding to the factorizations. By the lemma there is an infinite path. We find

  $$(a_0) \subsetneq (a_1) \subsetneq (a_2) \subsetneq ....$$

  the union I is an ideal, so of the form (i). But then $i \in a_n$ for some n, contradicting $(a_{n+1}) \neq (a_n)$. Thus the tree has finitely many vertices, corresponding to a finite factorization to irreducibles.

- Second, we show any factorization is unique. Suppose that

$$p_1 ... p_m = q_1 ... q_n$$

for irreducibles p_i, q_j(repetitions allowed). We wish to show that up to units we can match both sides (imagine

$$2 \cdot 3 = -2 \cdot -3$$

).

- Thus $p_1 | q_1 .... q_n$, so $p_1 | q_i$ for some i, but then $p_1 = q_i$ up to a unit, and we may cancel them out and continue.

2.0.3 Proposition- aposteriori

- Let R be a PID, then a, b don't share a prime divisor iff they are coprime (i.e., $\alpha a + \beta b = 1$).
- **Proof:**
- If $\alpha a + \beta b$, then any $p | a, b$ is a unit
- If a, b don't share a prime divisor, then $(a, b) = (i)$ since R is a PID, and i must be a unit otherwise a, b would share a divisor.

3 Solve exercises $7, 8, 9$

Part II
Review of algebraic integers

4 Number fields

4.0.1 Definition

A number field is a field $K/\mathbb{Q}$ of finite degree.

4.0.2 Remark

Concretely, every such K is of the form $K \cong \mathbb{Q}[\alpha]/p(\alpha)$ and so carries an embedding $K \hookrightarrow \mathbb{C}$.

4.1 Examples

1. $\mathbb{Q}[\omega_p]$ for $\omega = e^{2\pi i/p}$. **Spoiler:** is of degree $p-1$.
2. $\mathbb{Q}[\omega_m]$ for $\omega = e^{2\pi i/m}$. **Spoiler:** is of degree $\phi(m)$.
3. $\mathbb{Q}[\sqrt{m}]$ for $m \in \mathbb{Z}_{>0}$ squarefree are called the real quadratic fields (As they embed in $\mathbb{R}$).
4. $\mathbb{Q}[\sqrt{m}]$ for $m \in \mathbb{Z}_{<0}$ are the imaginary quadratic fields.

5 Algebraic integers

- **Definition:** Let $i : R \to S$ be a ring extension (in our case, $\mathbb{Z} \hookrightarrow K$ for a number field K), then $s \in S$ is algebraic over R if $\exists r_0, ..., r_{n-1}$with

$$s^n + i(r_{n-1})s^{n-1} + ... + i(r_0) = 0$$

- When $R \hookrightarrow S$ we can ignore i and abuse notation as $R \subset S$.

5.0.1 Definition

- $c_n x^n + ... + c_0 \in \mathbb{Z}[x]$ is primitive if $gcd(c_i) = 1$.
- Equivalently, f is primitive if for all primes p, $\bar{f} = 0$ (where $\bar{f}$ is the reduction mod p).

5.0.2 Gauss's lemma

- If f, g are primitive, so is $h = f \cdot g$.
- **Proof:** Suppose by contradiction $\bar{h} = 0$, then $\bar{f} \cdot \bar{g} = 0$, but $\mathbb{F}_p[x]$ is an integral domain.

5.0.3 Remark

- Gauss's lemma holds when replacing $\mathbb{Z}$ via any PID R, redefining primitive appropriately.

5.0.4 Proposition

- For R a PID, and $R \hookrightarrow Frac(R) \hookrightarrow S$ (think of $\mathbb{Z} \hookrightarrow K$ a number field), $s \in S$ is algebraic iff its minimal polynomial $p(x) \in Frac(R)[x]$, normalized to be monic, has coefficients in R.
- **Proof:** If the minimal polynomial $p(x)$ of s satisfies that, then s is algebraic over R. Conversely, if $\exists q(x)$ with $q(s) = 0$ and q monic with coefficients in R, then

$$q(x) = p(x)h(x)$$

for $h(x) \in Frac(R)[x]$. We claim that $p(x), h(x) \in R[x]$, which implies $p(x)$ is monic. Indeed by taking the lcm of the reduced denominators of the coefficients, we may renormalize $\tilde{p}(x) = p(x) \cdot l, \tilde{h}(x) \cdot l'$, $l, l' \in R$ s.t $\tilde{p(x)}, \tilde{h}(x)$ are primitive. Thus

$$l \cdot l' \cdot q(x) = \tilde{p} \cdot \tilde{h}$$

. Since the right hand side is primitive, $l \cdot l'$ is a unit, so l, l' are units, and thus $p(x), h(x) \in R[x]$.

5.0.5 Corollary

- $\mathbb{Z} \subset \mathbb{Q}$ are precisely the algebraic integers in $\mathbb{Q}$.